

Lab08 – HPA

1. Install the metrics server to actual resource metrics. You can find [installation instructions](<https://github.com/kubernetes-sigs/metrics-server#installation>) on the project's GitHub page.

```

vagrant@kubernetes-control-plane:~$ kubectl apply -f https://github.com/kubernetes-sigs/metrics-server/releases/latest/download/components.yaml
serviceaccount/metrics-server created
clusterrole.rbac.authorization.k8s.io/system:aggregated-metrics-reader created
clusterrole.rbac.authorization.k8s.io/system:metrics-server created
rolebinding.rbac.authorization.k8s.io/metrics-server-auth-reader created
clusterrolebinding.rbac.authorization.k8s.io/metrics-server:system:auth-delegator created
clusterrolebinding.rbac.authorization.k8s.io/system:metrics-server created
service/metrics-server created
deployment.apps/metrics-server created
apiservice.apiregistration.k8s.io/v1beta1.metrics.k8s.io created
vagrant@kubernetes-control-plane:~$ kubectl get po -n kube-system
NAME                                READY   STATUS    RESTARTS   AGE
calico-kube-controllers-6879d4fcdc-q9pvp    1/1     Running   0           18m
calico-node-4qb26                          1/1     Running   0           18m
calico-node-wdvwv                          1/1     Running   0           9m46s
coredns-6f6b679f8f-pxwt8                  1/1     Running   0           19m
coredns-6f6b679f8f-tw2fc                  1/1     Running   0           19m
etcd-kube-control-plane                   1/1     Running   0           19m
kube-apiserver-kube-control-plane          1/1     Running   0           19m
kube-controller-manager-kube-control-plane 1/1     Running   0           19m
kube-proxy-wcwhd                          1/1     Running   0           19m
kube-proxy-wqc6r                          1/1     Running   0           9m46s
kube-scheduler-kube-control-plane          1/1     Running   0           19m
metrics-server-548b7d8875-hjqx             0/1     Running   0           41s
vagrant@kubernetes-control-plane:~$

```

2. Create a Deployment named `nginx:1.23.4` with 1 replica. The Pod template of the Deployment should use container image `nginx:1.23.4`, set the CPU resource request to 0.5, and the memory resource request/limit to 500Mi.

```

vagrant@kubernetes-control-plane:~/Lab$ vim nginx-deployment.yaml
vagrant@kubernetes-control-plane:~/Lab$ cat nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx
spec:
  replicas: 1
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - image: nginx:1.23.4
          name: nginx
          resources:
            requests:
              cpu: "0.5"
              memory: "500Mi"
            limits:
              memory: "500Mi"
vagrant@kubernetes-control-plane:~/Lab$ _

```

```

vagrant@kubernetes-control-plane:~/Lab$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx created
vagrant@kubernetes-control-plane:~/Lab$
vagrant@kubernetes-control-plane:~/Lab$ kubectl get deploy,pod
NAME                                READY  UP-TO-DATE  AVAILABLE  AGE
deployment.apps/nginx              0/1    1           0          15s

NAME                                READY  STATUS              RESTARTS  AGE
pod/nginx-5bfd74dd7-jjz8h          0/1    ContainerCreating   0          15s
vagrant@kubernetes-control-plane:~/Lab$

```

3. Create a HorizontalPodAutoscaler for the Deployment named `nginx-hpa` that scales to minimum of 3 and a maximum of 8 replicas. Scaling should happen based on an average CPU utilization of 75%, and an average memory utilization of 60%.

```

vagrant@kubernetes-control-plane:~/Lab$ vim nginx-hpa.yaml
vagrant@kubernetes-control-plane:~/Lab$ cat nginx-hpa.yaml
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: nginx-hpa
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: nginx
  minReplicas: 3
  maxReplicas: 8
  metrics:
  - type: Resource
    resource:
      name: cpu
      target:
        type: Utilization
        averageUtilization: 75
  - type: Resource
    resource:
      name: memory
      target:
        type: Utilization
        averageUtilization: 60
vagrant@kubernetes-control-plane:~/Lab$
vagrant@kubernetes-control-plane:~/Lab$ kubectl apply -f nginx-hpa.yaml
horizontalpodautoscaler.autoscaling/nginx-hpa created
vagrant@kubernetes-control-plane:~/Lab$

```

4. Inspect the HorizontalPodAutoscaler object and identify the currently-utilized resources. How many replicas do you expect to exist?

```

vagrant@kubernetes-control-plane:~/Lab$ kubectl get hpa,deploy,pod
NAME                                REFERENCE                                TARGETS                                MINPODS
horizontalpodautoscaler.autoscaling/nginx-hpa  Deployment/nginx  cpu: 0%/75%, memory: 0%/60%  3

NAME                                READY  UP-TO-DATE  AVAILABLE  AGE
deployment.apps/nginx              3/3    3           3          33m

NAME                                READY  STATUS              RESTARTS  AGE
pod/nginx-5bfd74dd7-25s99          1/1    Running         1 (16m ago)  28m
pod/nginx-5bfd74dd7-9kt2d          1/1    Running         1 (16m ago)  28m
pod/nginx-5bfd74dd7-jjz8h          1/1    Running         1 (16m ago)  33m
vagrant@kubernetes-control-plane:~/Lab$

```